Aryan Naik

aryannaik1205@gmail.com github.com/RobotsAndAryan MEng Robotics and Artificial Intelligence, University College London, Expected 2029

EDUCATION

MEng Robotics and Artificial Intelligence — University College London (UCL) 2nd Year | Expected Graduation: 2029

PROJECTS

Aerospace Avionics Firmware — Student Rocketry Telemetry Payload

- Delivered autonomous parachute deployment under safety-critical flight conditions by implementing a finite state machine governing phase transitions across ascent, apogee, and descent, eliminating the need for manual recovery intervention across all test flight sequences
- Achieved continuous IMU inertial measurement unit data acquisition and altitude sensing by integrating multiple sensors over I2C serial bus on a dual-MCU embedded architecture with interrupt-driven real-time execution, sustaining uninterrupted sensor polling throughout powered flight
- Enabled post-flight data recovery and live ground monitoring by implementing onboard flash data logging and wireless telemetry transmission, producing a complete retrievable dataset for every flight event recorded during testing

Hardware Robotics Integration — Competition Platform

- Delivered a fully operational robotic platform that completed all assigned competition tasks by integrating microcontrollers, motor drivers, actuators, ultrasonic and IR distance sensors, and power electronics onto a breadboard circuit assembled from schematic to physical build
- Reduced integration faults to zero prior to competition deployment by applying systematic fault-finding and hardware debugging across signal lines, power rails, and sensor outputs to verify signal integrity at each assembly stage
- Produced structural enclosures and mounting geometry meeting assembly fit requirements by completing mechanical CAD design in Autodesk Fusion 360, with components verified through rapid prototyping and 3D-printed acceptance builds

SKILLS

Embedded and Electrical Engineering C++, interrupt-driven real-time firmware, dual-MCU embedded architecture, I2C, SPI, UART serial communication, sensor fusion, signal conditioning, power budgeting, voltage regulation, H-bridge and servo motor control, PCB layout concepts, breadboard electronics design, component selection

Mechanical and Systems Engineering Autodesk Fusion 360 CAD, structural design, tolerancing, rapid prototyping, 3D printing, hardware-software co-design, systems integration, acceptance testing

Aerospace and Defence-Relevant Domains Aerospace avionics, safety-critical embedded software, rocketry telemetry payload, flight phase state machine, autonomous recovery systems, defence electronics

Platforms and Tools Git, GitHub

ADDITIONAL INFORMATION

- Eligible to work in the UK
- Available for summer placement 2027
- Locations considered: Bolton, Bristol, Stevenage

Aryan Naik aryannaik1205@gmail.com github.com/RobotsAndAryan

Hiring Team MBDA Bolton / Bristol / Stevenage

Dear Hiring Team,

I am writing to apply for the Summer Placement 2027 at MBDA. As a second-year MEng Robotics and Artificial Intelligence student at University College London, I am drawn specifically to MBDA's work at the intersection of embedded systems engineering and safety-critical defence applications. MBDA's role in delivering sovereign missile and advanced weapon capabilities to allied armed forces, and its cross-border integration of engineering expertise across air, land, and naval domains, represents exactly the kind of technically demanding, high-consequence environment in which I want to develop as an engineer.

My most directly relevant experience is the ALRS 007 Capsule project, an aerospace avionics firmware system I developed for a student rocketry telemetry payload. Working on a dual-MCU embedded architecture, I wrote interrupt-driven C++ firmware integrating IMU, altitude, and acceleration sensors over I2C, implementing sensor fusion for real-time attitude estimation. I designed a finite state machine to govern autonomous parachute deployment under safety-critical flight conditions, alongside wireless telemetry transmission and onboard flash data logging. This project required me to balance hardware-software co-design, power budgeting, and signal integrity under strict reliability constraints, producing a functional flight computer capable of autonomous recovery decisions in a live launch environment.

At MBDA I would contribute embedded systems and real-time firmware skills directly applicable to avionics, guidance, and defence electronics work. I am keen to deepen my understanding of how safety-critical software is engineered and verified at the standards demanded by operational defence systems, and to learn how large-scale, cross-disciplinary engineering programmes are structured and delivered within a leading defence organisation.

Yours sincerely, Aryan Naik